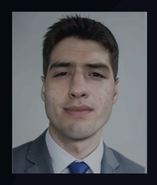
DREAM-TO-WORLD ENGINE:

GENERATING IMMERSIVE 3D WORLDS



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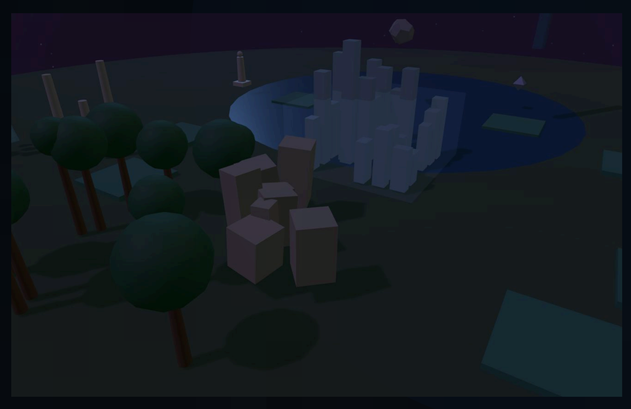
PROJECT OVERVIEW

This project transforms natural language descriptions into interactive 3D environments.

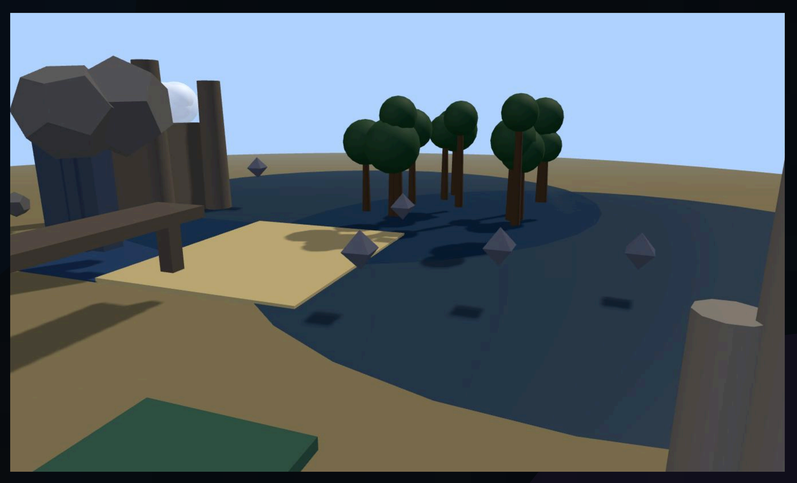
Users input a dream or idea, which is parsed into structured entities and relationships.

These are then used to procedurally generate a navigable world in real time, creating a direct pipeline from imagination to visualization.



SYSTEM ARCHITECTURE

GENERATED 3D WORLD



DREAM PROMPT

"A SUNNY BEACH AT THE OCEAN

WITH WAVES AND GOLDEN SAND,
TURTLES IN SHALLOW WATER,
SHARKS IN THE DEEP SEA,

AND A PLANE FLYING IN THE SKY"

KEY FEATURES

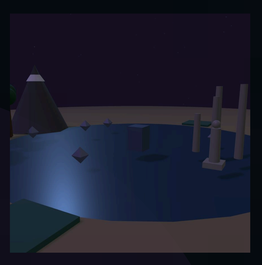
· Natural language to structured world conversion

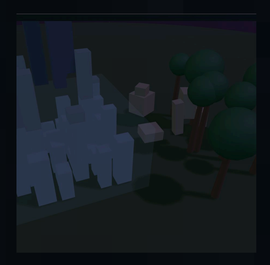
· Procedural terrain and object generation

· Entity and relationship-based scene construction

· Dynamic scene composition from minimal input

· Extensible system for adding new entity types and behaviors





CHALLENGES

· Handling ambiguous or incomplete user input

· Balancing scene realism with performance constraints

· Designing a scalable entity classification system

FUTURE WORK

· Expand entity recognition to include animals and complex objects

· Improve environmental detail and realism

· Enhance real-time scene generation and rendering

RESULTS & ACHIEVEMENTS

· Successfully generated coherent 3D scenes from simple text prompts

· Built a working entity-relationship pipeline for world modeling

· Achieved real-time scene generation and rendering